

	0.010 V – 0.02 V
7(a)	The magnitude of the induced <i>emf</i> in a conductor is proportional to the rate of change with time t of the magnetic flux Φ that cuts across the conductor.
7(b)	The curved wall of tube A will experience a changing magnetic flux as the magnetised steel block falls through it, which induces an emf and hence a current, according to Faraday's Law. The magnetic flux is decreasing above the magnet and increasing below the falling magnet. By Lenz's law, the induced current will flow such that it will prevent the magnetic flux from changing. As a result, this induced current produces magnetic field that attracts the magnet above and repel the magnet below it, generating a net magnetic force upward. The total force acting on the magnet for tube A is therefore much smaller compared to tube B, the net downward acceleration is therefore much smaller and hence take a longer time to fall.

Qn	Solution
$g(a)(i)$	